In [ ]:
```
 Experiment 3: Creating DataFrames & Analyzing Properties

PIN:-241UDS0101

 Aim:-
Create Pandas DataFrames using NumPy arrays and CSV files, then analyze
DataFrame properties and descriptive statistics.
```

In [ ]:
```
## Step 1: Install and Import Libraries
```

In [6]:
```
import numpy as np
import pandas as pd
```

In [ ]:
```
## Step 2: Create a 2-D NumPy Array
Rows are branches and columns are weeks.
```

In [8]:
```
sales_241UDS0101 = np.array([[342, 510, 684, 825], [427, 672, 291, 648],
[864, 379, 592, 746]])
sales_241UDS0101
```

Out[8]:
```
array([[342, 510, 684, 825],
       [427, 672, 291, 648],
       [864, 379, 592, 746]])
```

In [9]:
```
sales_241UDS0101.shape, sales_241UDS0101.ndim
```

Out[9]:
```
((3, 4), 2)
```

In [10]:
```
df_index_241UDS0101 = ['Branch_A', 'Branch_B', 'Branch_C']
df_columns_241UDS0101 = ['Week_1', 'Week_2', 'Week_3', 'Week_4']
df_241UDS0101 = pd.DataFrame(data=sales_241UDS0101,
index=df_index_241UDS0101, columns=df_columns_241UDS0101)
df_241UDS0101
```

Out[10]:

	Week_1	Week_2	Week_3	Week_4
Branch_A	342	510	684	825
Branch_B	427	672	291	648
Branch_C	864	379	592	746

In [ ]:
```
df_241UDS0101['Week_5'] = [305, 329, 418]
df_241UDS0101
```

In [11]:
```
branch_d_241UDS0101 = pd.DataFrame({'Week_1':[172], 'Week_2':[806],
'Week_3':[314], 'Week_4':[358], 'Week_5':[549]}, index=['Branch_D'])
df_241UDS0101 = pd.concat([df_241UDS0101, branch_d_241UDS0101])
df_241UDS0101
```

Out[11]:

	Week_1	Week_2	Week_3	Week_4	Week_5
Branch_A	342	510	684	825	NaN
Branch_B	427	672	291	648	NaN
Branch_C	864	379	592	746	NaN
Branch_D	172	806	314	358	549.0

In [ ]:
```
## Step 4: Create a DataFrame from a CSV File
Keep the CSV file in the same folder as this notebook.
```

In [12]:
```
csv_path_241UDS0101 = 'Experiment_3_241UDS0101_Student_Data.csv'
stu_df_241UDS0101 = pd.read_csv(csv_path_241UDS0101)
stu_df_241UDS0101.head()
```

Out[12]: Columns 1–8 of 22

	school	sex	age	address	famsize	Pstatus	Medu
0	MS	M	21	R	GT3	A	3
1	MS	M	21	R	LE3	A	0
2	GP	M	16	R	GT3	T	1
3	MS	M	18	R	LE3	T	1
4	GP	M	20	R	LE3	T	4

Columns 9–15 of 22 · first column repeated

	Fedu	Mjob	Fjob	...	internet	romantic	famrel
0	0	other	other	...	no	no	1
1	2	services	other	...	yes	yes	4
2	3	teacher	at_home	...	yes	yes	1
3	0	services	health	...	yes	yes	1
4	4	teacher	teacher	...	no	no	1

Columns 16–22 of 22 · first column repeated

	freetime	goout	Dalc	Walc	health	absences	passed
0	4	4	1	5	2	6	yes
1	5	4	2	4	3	3	yes
2	5	4	2	1	2	12	no
3	5	5	3	3	2	7	yes
4	4	2	2	5	4	4	no

5 rows × 31 columns

In [ ]:
```
## Step 5: Analyze Important DataFrame Properties
```

In [13]:
```
stu_df_241UDS0101.head()
```

Out[13]: Columns 1–8 of 22

	school	sex	age	address	famsize	Pstatus	Medu
0	MS	M	21	R	GT3	A	3
1	MS	M	21	R	LE3	A	0
2	GP	M	16	R	GT3	T	1
3	MS	M	18	R	LE3	T	1
4	GP	M	20	R	LE3	T	4

Columns 9–15 of 22 · first column repeated

	Fedu	Mjob	Fjob	...	internet	romantic	famrel
0	0	other	other	...	no	no	1
1	2	services	other	...	yes	yes	4
2	3	teacher	at_home	...	yes	yes	1
3	0	services	health	...	yes	yes	1
4	4	teacher	teacher	...	no	no	1

Columns 16–22 of 22 · first column repeated

	freetime	goout	Dalc	Walc	health	absences	passed
0	4	4	1	5	2	6	yes
1	5	4	2	4	3	3	yes
2	5	4	2	1	2	12	no
3	5	5	3	3	2	7	yes
4	4	2	2	5	4	4	no

5 rows × 31 columns

In [14]:
```
stu_df_241UDS0101.shape
```

Out[14]:
```
(24, 31)
```

In [15]:
```
stu_df_241UDS0101.index
```

Out[15]:
```
RangeIndex(start=0, stop=24, step=1)
```

In [16]:
```
stu_df_241UDS0101.dtypes
```

Out[16]:
```
school        object
sex           object
age            int64
address       object
famsize       object
Pstatus       object
Medu           int64
Fedu           int64
Mjob          object
Fjob          object
reason        object
guardian      object
traveltime     int64
studytime      int64
failures       int64
schoolsup     object
famsup        object
paid          object
activities    object
nursery       object
higher        object
internet      object
romantic      object
famrel         int64
freetime       int64
goout          int64
Dalc           int64
Walc           int64
health         int64
absences       int64
passed        object
dtype: object
```

In [17]:
```
stu_df_241UDS0101.columns
```

Out[17]:
```
Index(['school', 'sex', 'age', 'address', 'famsize', 'Pstatus', 'Medu',
'Fedu',
       'Mjob', 'Fjob', 'reason', 'guardian', 'traveltime', 'studytime',
       'failures', 'schoolsup', 'famsup', 'paid', 'activities', 'nursery',
       'higher', 'internet', 'romantic', 'famrel', 'freetime', 'goout',
'Dalc',
       'Walc', 'health', 'absences', 'passed'],
      dtype='object')
```

In [ ]:
```
## Step 6: Information About the DataFrame
info() summarizes column names, non-null counts, data types, and memory use.
```

In [18]:
```
stu_df_241UDS0101.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 24 entries, 0 to 23
Data columns (total 31 columns):
 #   Column      Non-Null Count  Dtype 
---  ------      --------------  ----- 
 0   school      24 non-null     object
 1   sex         24 non-null     object
 2   age         24 non-null     int64 
 3   address     24 non-null     object
 4   famsize     24 non-null     object
 5   Pstatus     24 non-null     object
 6   Medu        24 non-null     int64 
 7   Fedu        24 non-null     int64 
 8   Mjob        24 non-null     object
 9   Fjob        24 non-null     object
 10  reason      24 non-null     object
 11  guardian    24 non-null     object
 12  traveltime  24 non-null     int64 
 13  studytime   24 non-null     int64 
 14  failures    24 non-null     int64 
 15  schoolsup   24 non-null     object
 16  famsup      24 non-null     object
 17  paid        24 non-null     object
 18  activities  24 non-null     object
 19  nursery     24 non-null     object
 20  higher      24 non-null     object
 21  internet    24 non-null     object
 22  romantic    24 non-null     object
 23  famrel      24 non-null     int64 
 24  freetime    24 non-null     int64 
 25  goout       24 non-null     int64 
 26  Dalc        24 non-null     int64 
 27  Walc        24 non-null     int64 
 28  health      24 non-null     int64 
 29  absences    24 non-null     int64 
 30  passed      24 non-null     object
dtypes: int64(13), object(18)
memory usage: 5.9+ KB
```

In [ ]:
```
## Step 7: Descriptive Statistics
describe() summarizes numerical columns by default.
```

In [19]:
```
stu_df_241UDS0101.describe()
```

Out[19]:

Columns 1–8 of 14

	age	Medu	Fedu	traveltime	studytime	failures	famrel
count	24.000000	24.000000	24.000000	24.000000	24.000000	24.000000	24.000000
mean	18.000000	1.875000	1.916667	2.583333	2.375000	1.000000	2.375000
std	2.226325	1.454005	1.442120	1.059806	1.055524	0.932505	1.438976
min	15.000000	0.000000	0.000000	1.000000	1.000000	0.000000	1.000000
25%	16.000000	0.750000	0.750000	2.000000	1.750000	0.000000	1.000000
50%	18.000000	2.000000	2.000000	3.000000	2.000000	1.000000	2.000000
75%	20.000000	3.000000	3.000000	3.000000	3.000000	2.000000	3.250000
max	21.000000	4.000000	4.000000	4.000000	4.000000	3.000000	5.000000

Columns 9–14 of 14 · first column repeated

	freetime	goout	Dalc	Walc	health	absences
count	24.000000	24.000000	24.000000	24.000000	24.000000	24.000000
mean	3.458333	2.958333	3.166667	2.875000	2.958333	6.583333
std	1.215092	1.458980	1.578846	1.512628	1.160179	4.200587
min	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000
25%	2.000000	2.000000	2.000000	1.000000	2.000000	3.000000
50%	4.000000	3.000000	3.000000	3.000000	3.000000	6.500000
75%	4.000000	4.000000	5.000000	4.000000	3.250000	10.000000
max	5.000000	5.000000	5.000000	5.000000	5.000000	14.000000

In [ ]:
```
## Step 8: Transpose Rows and Columns
```

In [21]:
```
stu_df_241UDS0101.head(10).transpose()
```

Out[21]: Columns 1–8 of 11

	0	1	2	3	4	5	6
school	MS	MS	GP	MS	GP	GP	MS
sex	M	M	M	M	M	F	F
age	21	21	16	18	20	18	15
address	R	R	R	R	R	R	R
famsize	GT3	LE3	GT3	LE3	LE3	LE3	GT3
Pstatus	A	A	T	T	T	A	A
Medu	3	0	1	1	4	0	2
Fedu	0	2	3	0	4	0	3
Mjob	other	services	teacher	services	teacher	health	at_home
Fjob	other	other	at_home	health	teacher	teacher	other
reason	other	home	home	other	reputation	reputation	other
guardian	father	father	father	father	other	other	mother
traveltime	4	3	1	2	2	1	4
studytime	1	2	3	2	3	2	4
failures	0	0	0	2	0	1	2
schoolsup	no	no	no	no	yes	yes	no
famsup	no	yes	no	no	yes	no	yes
paid	no	no	yes	yes	yes	no	no
activities	yes	yes	yes	no	no	no	yes
nursery	no	yes	no	no	yes	no	yes
higher	yes	no	yes	no	no	no	no
internet	no	yes	yes	yes	no	yes	yes
romantic	no	yes	yes	yes	no	yes	yes
famrel	1	4	1	1	1	1	2
freetime	4	5	5	5	4	4	2
goout	4	4	4	5	2	3	2
Dalc	1	2	2	3	2	5	4
Walc	5	4	1	3	5	2	4
health	2	3	2	2	4	3	5
absences	6	3	12	7	4	3	1
passed	yes	yes	no	yes	no	no	no

Columns 9–11 of 11 · first column repeated

	7	8	9
school	GP	MS	MS
sex	F	M	F
age	20	20	18

	7	8	9
address	U	R	R
famsize	GT3	GT3	LE3
Pstatus	T	T	A
Medu	2	2	4
Fedu	2	4	2
Mjob	teacher	services	at_home
Fjob	services	services	services
reason	home	course	home
guardian	mother	mother	father
traveltime	2	3	3
studytime	1	4	1
failures	1	2	1
schoolsup	yes	yes	no
famsup	no	no	no
paid	no	yes	no
activities	no	yes	yes
nursery	no	yes	no
higher	yes	yes	no
internet	no	yes	no
romantic	no	no	yes
famrel	1	5	2
freetime	4	3	2
goout	1	1	5
Dalc	5	4	1
Walc	5	1	1
health	1	2	2
absences	1	4	10
passed	yes	no	yes

In [ ]:
```
Result :-
DataFrames were created from a NumPy array and a CSV file. Shape, index,
data types, column names, DataFrame information, descriptive statistics, and
transposed data were analyzed.
```